Iterative Prompting

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected prompt-revision record.

Explain the comparison, trace paired outcomes, audit limits and design a next check.

Keep one habit: Retain prompt versions, source/rubric, per-case outcomes, regressions and fresh-case plans.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Examples and **Rubric**s. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Sampling and Variability. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Predictive and Generative AI. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Examples and Rubrics

Sampling and Variability

Predictive and Generative AI

W1 · Explain a controlled iteration

> Name the diagnosed failure, one change and fixed conditions.

Your evidence record

- Six-case W2 review: P passes1,2,3,4;Q passes1,2,3,5,6;missing-approval slice4,6 gives1/2 each but different passing IDs. Separate four-question pictured/lab record: P passes1,2;Q passes1,2,3;case4stillfails. Proposed R is not run. These records use different case sets and must not share a denominator.

- Reference for this exercise: open this week's student lesson, "Iterative Prompting", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the separate six-case record

> Keep total and missing-approval slice distinct.

Your evidence record

- Six-case W2 review: P passes1,2,3,4;Q passes1,2,3,5,6;missing-approval slice4,6 gives1/2 each but different passing IDs. Separate four-question pictured/lab record: P passes1,2;Q passes1,2,3;case4stillfails. Proposed R is not run. These records use different case sets and must not share a denominator.

- On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Audit the evidence boundary

> A selected development result is not untouched final performance.

Your evidence record

- Six-case W2 review: P passes1,2,3,4;Q passes1,2,3,5,6;missing-approval slice4,6 gives1/2 each but different passing IDs. Separate four-question pictured/lab record: P passes1,2;Q passes1,2,3;case4stillfails. Proposed R is not run. These records use different case sets and must not share a denominator.

- A report omits this qualification: Prompt selection on development cases cannot be reported as independent final performance. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Design a revealing follow-up

> Record predictions and preserve unrun or unsuccessful outcomes.

Your evidence record

- Six-case W2 review: P passes1,2,3,4;Q passes1,2,3,5,6;missing-approval slice4,6 gives1/2 each but different passing IDs. Separate four-question pictured/lab record: P passes1,2;Q passes1,2,3;case4stillfails. Proposed R is not run. These records use different case sets and must not share a denominator.

- Reference for this exercise: open this week's student lesson, "Iterative Prompting", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Design a new format-revision evaluation independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Design the comparison; no revised outputs, source text or model scores are supplied. Fix the approved evidence, factual criteria, task cases and model/settings while changing the format instruction. Define three-bullet compliance separately from factual support, and preserve unrun outcomes as unknown.

A factual response uses four paragraphs when the contract requires three bullets. Design a controlled prompt revision and evaluation.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Changed independent case

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Maintain prompt **version**, diagnosed failure, change, and before/after **rubric** scores.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Trace the worked case for Controlled prompt experiments. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Controlled prompt experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A summary has accurate facts but uses unexplained terms. Propose one prompt change and a comparison that checks both accessibility and factual preservation.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A factually correct explanation is too difficult for a nine-year-old.

Make and test one focused revision.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

You change a prompt's audience, length and source rules together; the result improves. What can you conclude, and how would you investigate the cause?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 7 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A summary has accurate facts but uses unexplained terms. Propose one prompt change and a comparison that checks both accessibility and factual preservation.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Additional practice · Compare the older eight-case record

REFERENCE · SECTION 15

Additional practice · Compare the older eight-case record

> This retained arithmetic task is separate from the selected format-revision design.

Earlier task and saved response

- On eight **paired case**s A passes 1–5; B passes 1–4, 6, 7. Special cases are 5 and 8. Compare overall and special-case results; select by total passes.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

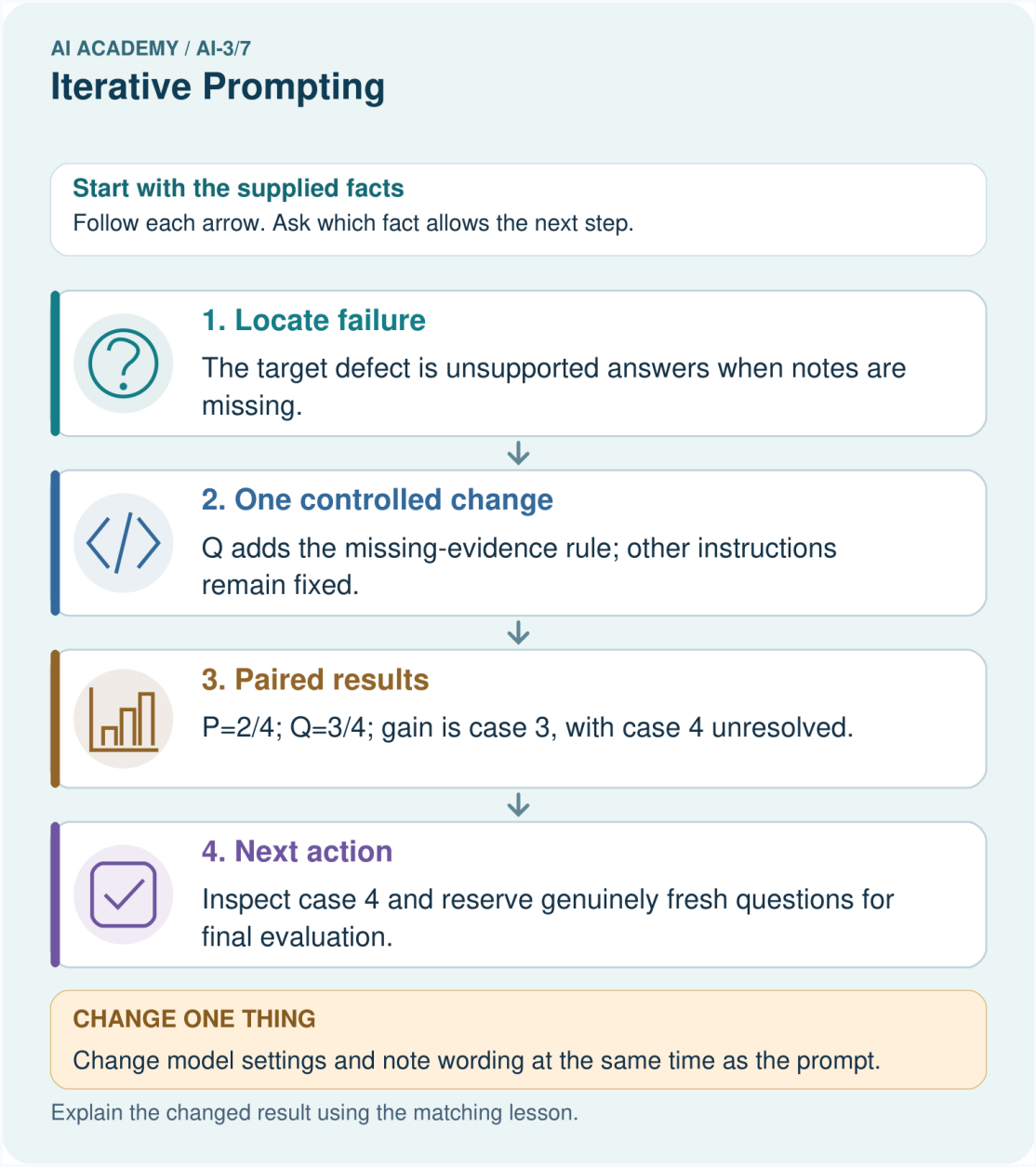
> Keep the explanation and the support you used.

Use this when helpful

A fictional Study Buddy prompt invents a definition when approved notes omit it.

P asks “answer helpfully”. Revision Q adds “if approved notes lack the answer, say not supplied”.

Hold model settings and four development questions fixed. P passes 1,2; Q passes 1,2,3. Case 4 still fails.



Follow the labelled picture with the matching explanation. Open the vector diagram

What **evidence** supports improvement here, and what remains unproven?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Maintain a Study Buddy experiment log linking failure, single revision and paired **rubric** results.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Verified Study Buddy

Current milestone: Make a transparent first **version**. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Maintain a Study Buddy experiment log linking failure, single revision and paired **rubric** results.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional Study Buddy prompt invents a definition when approved notes omit it.
- Which labelled step matches this detail? Inspect case 4 and reserve genuinely fresh questions for final evaluation.
- 1. Locate failure
- 2. Next action

- 3. One controlled change
- 4. Paired results
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Next action Inspect case 4 and reserve genuinely fresh questions for final evaluation. An **iteration** should respond to a recorded failure with a precise change.

Comparing the same questions reveals whether other behavior changed too. The invented results show a development gain of one case, not a guarantee that the new instruction always works.

Save prompt text, settings, source notes and case-level outcomes.

Changed-case question

- Changed-case practice: Change model settings and note wording at the same time as the prompt. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. The comparison no longer isolates the prompt revision.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The comparison no longer isolates the prompt revision. Several changes could explain the altered **output**s; record them or run separate controlled comparisons.

Repair and save your work

Review Examples and **Rubric**s (ai-3/6). Goal: Distinguish a format example from approved factual evidence. Short practice: Explain why neither original nor changed draft is a full pass, and provide a complete repaired card.. Return to this week and explain what you changed.

Review Sampling and Variability (ai-3/4). Goal: Sample from a distribution using explicit cumulative intervals. Short practice: Does T=0.5 make B impossible, and does choosing A guarantee factual accuracy?.

Return to this week and explain what you changed.

Review Predictive and Generative AI (ai-3/1). Goal: Distinguish discriminative **prediction** from generation by inspecting the requested **output**.

Short practice: Identify the first unsupported claim and explain what source would be needed before asserting it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

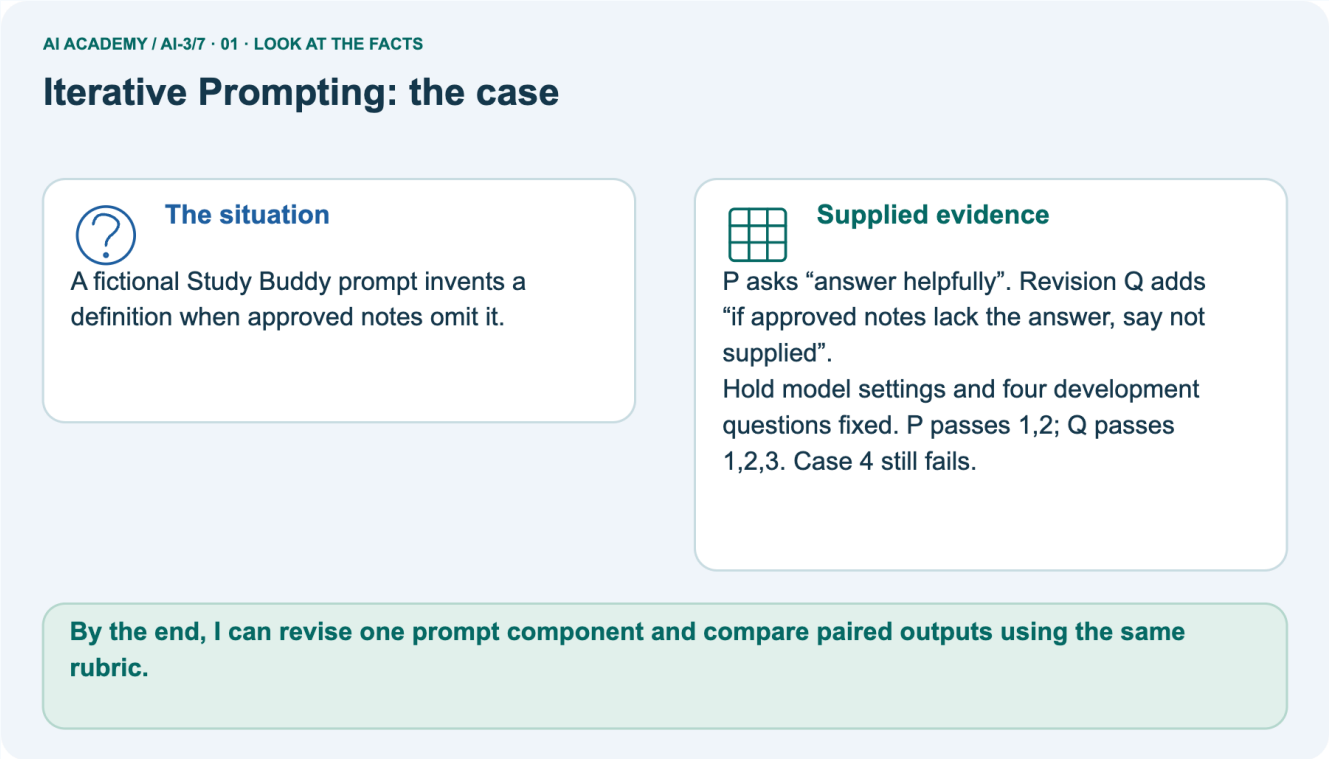
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Iterative Prompting: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

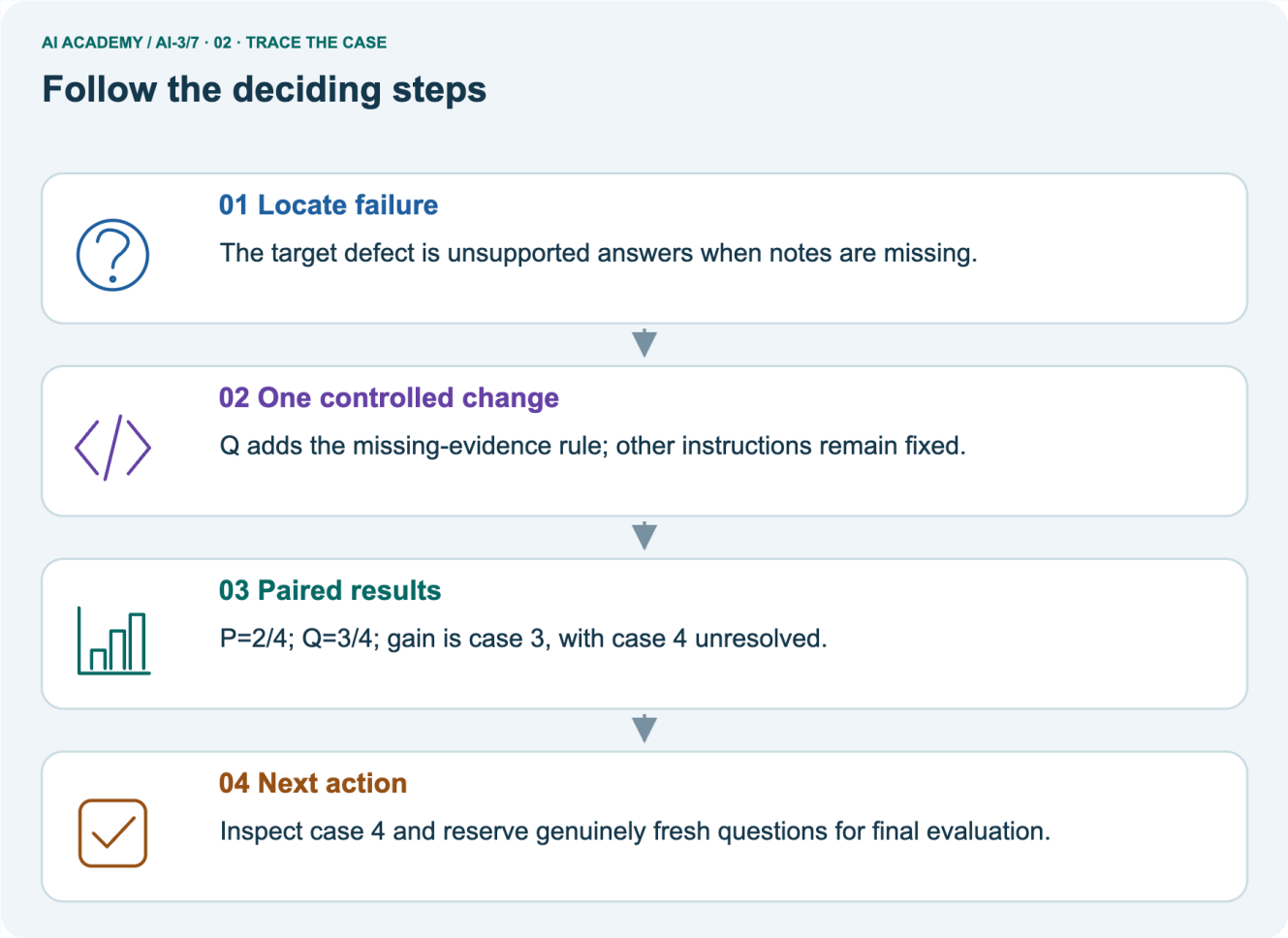
A fictional Study Buddy prompt invents a definition when approved notes omit it. Supplied facts: P asks “answer helpfully”. Revision Q adds “if approved notes lack the answer, say not supplied”.

Hold model settings and four development questions fixed. P passes 1,2; Q passes 1,2,3. Case 4 still fails.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



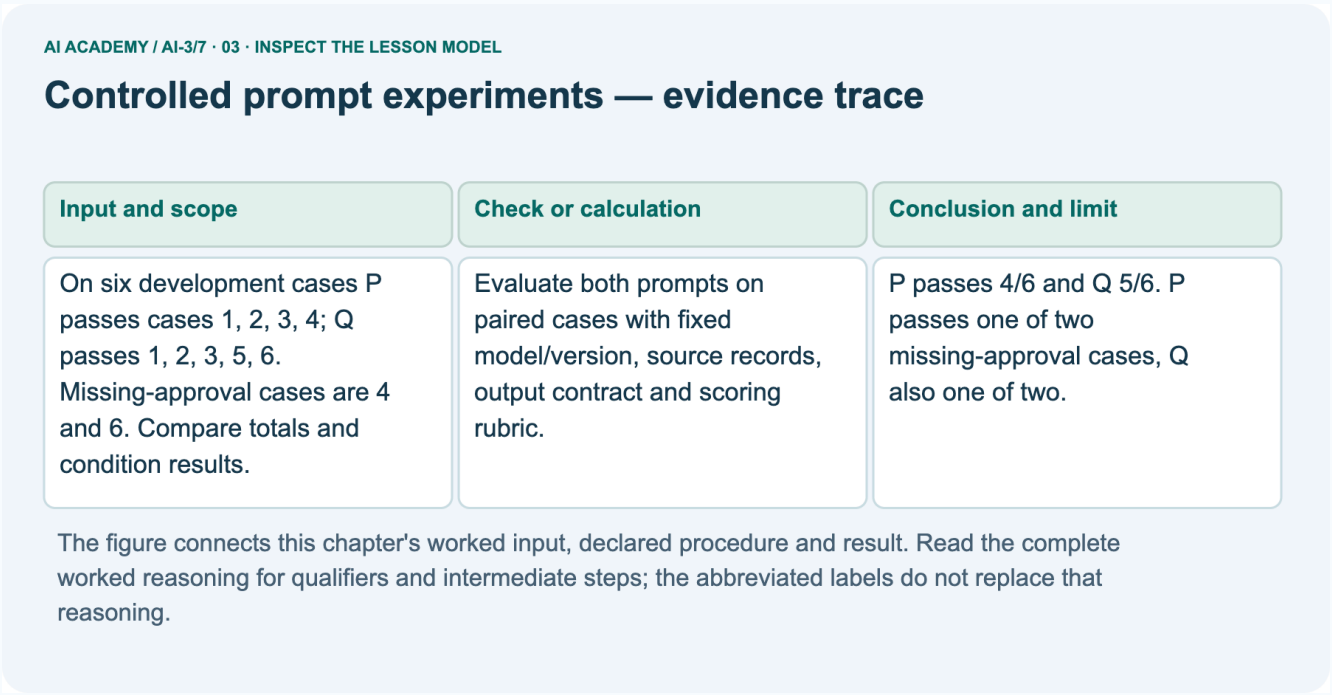
Follow the labelled picture with the matching explanation. Open the vector diagram

Locate failure: The target defect is unsupported answers when notes are missing. → One controlled change: Q adds the missing-**evidence** rule; other instructions remain fixed. → Paired results: P=2/4; Q=3/4; gain is case 3, with case 4 unresolved. → Next action: Inspect case 4 and reserve genuinely fresh questions for final evaluation.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Controlled prompt experiments — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Controlled prompt experiments — **evidence** trace. On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6.

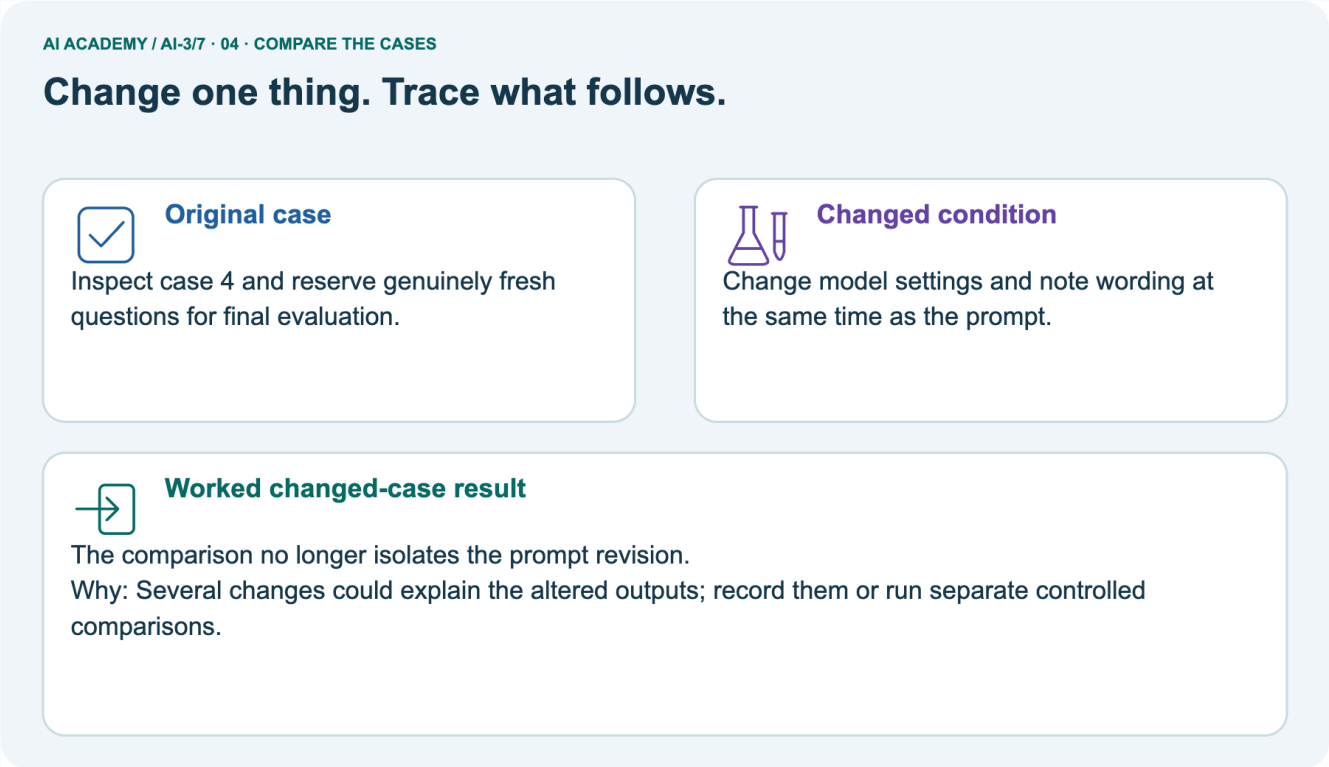
Compare totals and condition results. | Evaluate both prompts on **paired case**s with fixed model/**version**, source records, **output** contract and scoring **rubric**. | P passes 4/6 and Q 5/6.

P passes one of two missing-approval cases, Q also one of two.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

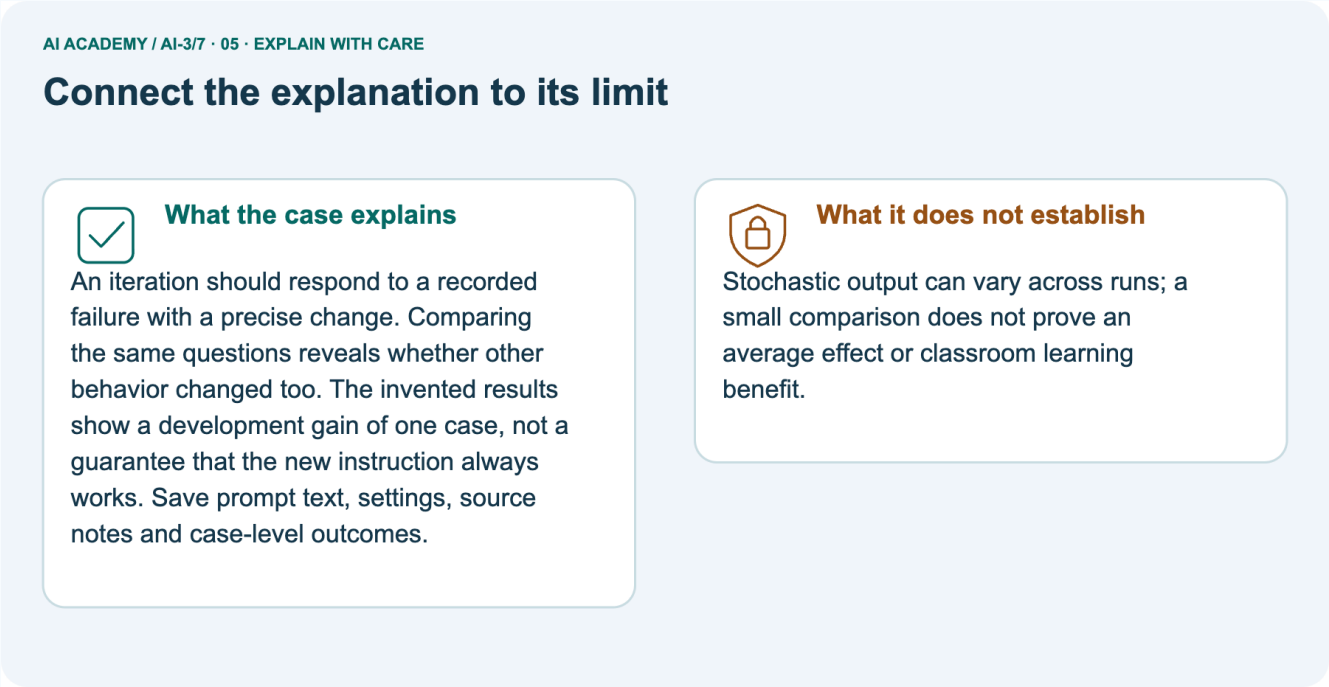
Original case: Inspect case 4 and reserve genuinely fresh questions for final evaluation. Changed condition: Change model settings and note wording at the same time as the prompt.

Expected result: The comparison no longer isolates the prompt revision. Reason: Several changes could explain the altered **output**s; record them or run separate controlled comparisons.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An **iteration** should respond to a recorded failure with a precise change. Comparing the same questions reveals whether other behavior changed too.

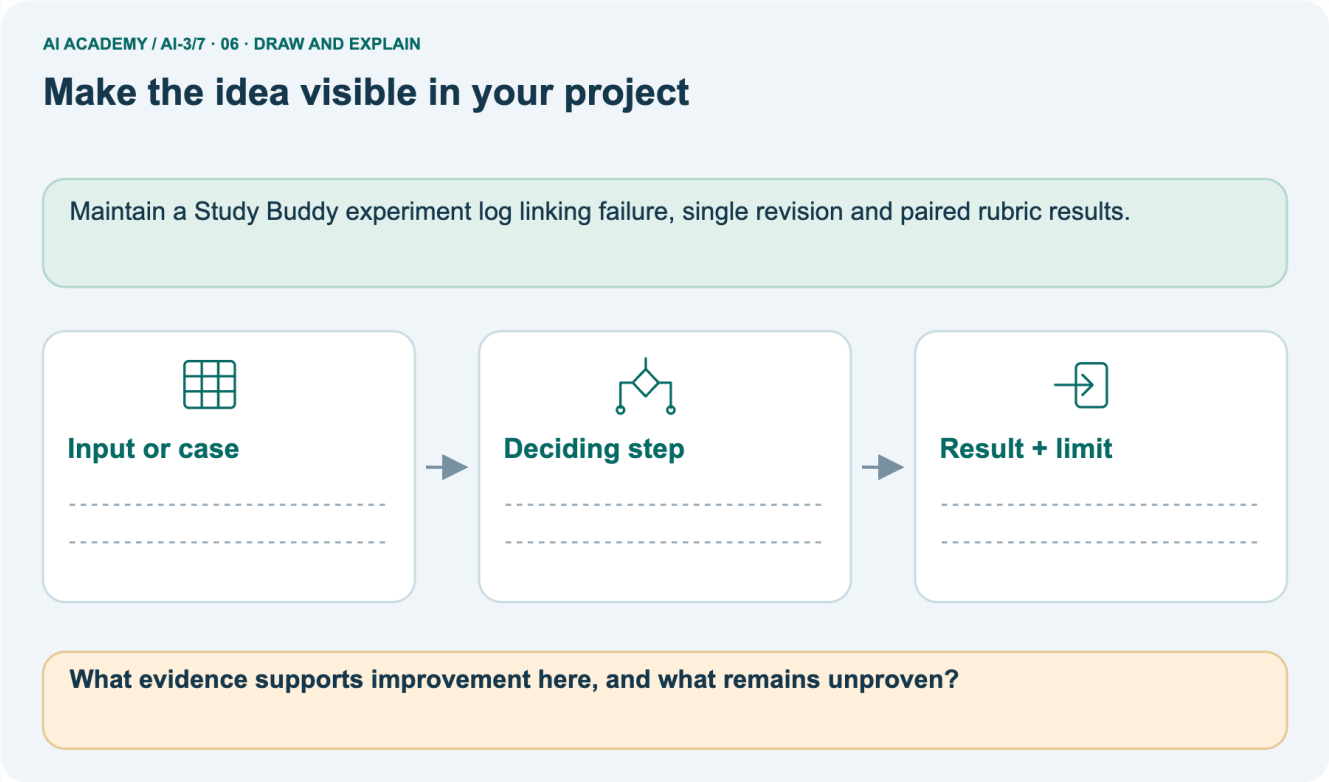
The invented results show a development gain of one case, not a guarantee that the new instruction always works. Save prompt text, settings, source notes and case-level outcomes.

Limit: Stochastic **output** can vary across runs; a small comparison does not prove an average effect or classroom learning benefit.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Maintain a Study Buddy experiment log linking failure, single revision and paired **rubric** results. Draw your **input**, deciding step, result and limit.

Supported question: What **evidence** supports improvement here, and what remains unproven?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Iterative Prompting: the case

02. Follow the deciding steps

03. Controlled prompt experiments — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.